Deep Learning versus a Non-Parametric Baseline for Freeway Traffic-Flow Anomaly Detection: A Uniform Protocol and Labeling-Sensitivity Design on PeMS-D4

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Abstract*—Deep learning detectors for traffic anomalies are usually reported under heterogeneous datasets, splits, thresholds and labels, while recent benchmark studies show that simple statistical detectors are often competitive once evaluation flaws are removed. We present a uniform protocol, whose main rules were fixed before any network result was seen, that compares six deep learning families (LSTM, GRU, CNN-1D, autoencoder, Transformer and a CNN-LSTM-GRU hybrid) and one published model (SI-DMSeq) against a non-parametric rolling-statistics baseline on PeMS-D4 freeway flow series. The protocol uses walk-forward temporal validation, a common reduced tuning procedure for the six trained families with the published model left at its default configuration, three seeds for the final models, PR-AUC without point adjustment, paired tests with Holm correction, latency benchmarking, and two separately constructed labeling schemes, a causal statistical proxy and the controlled injection of point, contextual and collective anomalies, to test whether the detector ranking depends on how anomalies are defined. Preliminary outputs, which passed the specified automated consistency checks, cover data curation (291 of 307 sensors retained), both labeling schemes and the baseline. The proxy scheme yields a prevalence of 0.10 percent and only 29.2 percent of evaluation blocks contain both classes; its positive set barely overlaps with the injected one (Jaccard index of at most 0.0012); and the baseline reaches a block-mean test PR-AUC of 0.345 under injection, about 11 times chance, but a pooled PR-AUC of only 0.003 under the proxy, about three times chance. Measured training times forced a reduced, pre-declared hyperparameter search. Deep learning results and statistical tests are pending.

***Index Terms*—anomaly detection, deep learning, evaluation protocol, PeMS, time series, traffic flow, synthetic anomaly injection**

I. Introduction

Context and who is affected. Traffic management centers, such as the California Department of Transportation (Caltrans), which sponsors PeMS [1], rely on freeway sensors to detect disruptions, and drivers bear the delays. The U.S. Federal Highway Administration attributes about half of congestion to temporary disruptions, with incidents alone accounting for 25% of congestion [2]. In the San Francisco Bay Area, INRIX estimated that congestion cost each driver 79 hours and USD 2 250 in 2017, the year before the period studied here [3]. Detecting atypical observations promptly therefore has operational value. If detectors are adopted on the strength of results that cannot be compared, an operator may pay for costly deep models without a demonstrated advantage over a simple rule, which is why a uniform evaluation protocol is needed.

Freeway loop detectors produce dense, seasonal and noisy flow series in which timely identification of atypical observations supports traffic monitoring. DL detectors based on recurrence, convolution, reconstruction or attention have been proposed for this task [4]–[7], but their results are hard to compare: studies differ in data, splits, labels, thresholds and metrics, and large public freeway datasets such as PeMS were built for forecasting and carry no anomaly labels [8]. Labels are therefore derived from incident logs, rules or simulation [9], [10], and a statistical anomaly is not necessarily a real incident.

Two lines of evidence motivate a stricter evaluation. Point adjustment (PA) inflates scores so much that random scores reach state-of-the-art F1 [11], and popular benchmarks are largely solvable by one-line moving-statistics rules [12]. When evaluation is controlled, classical detectors match or outperform deep ones on generic benchmarks [13]–[16] and on expert-labeled traffic volume [17]; moreover, label quality

changes conclusions in traffic incident detection [9], and rankings shift with anomaly type and metric [16], [18].

Research object and problem. The object of study is DL-based anomaly detection in freeway traffic flow, analyzed on PeMS-D4 with the detector × labeling scheme × sensor-fold block as unit of analysis. The problem is whether DL detectors provide a reproducible advantage over a rolling-statistics baseline, and whether the resulting ranking remains stable across operational definitions of anomaly and under an inference-cost constraint.

Knowledge gap. Uniform comparisons already exist for generic series [13], [16] and for urban traffic volume [17]. What is not known is whether the relative standing of DL detectors against a simple baseline on freeway flow survives a change in how anomalies are defined. In the literature we reviewed, no study examines this on one temporal partition of a freeway dataset while jointly comparing generic DL families, a non-parametric baseline and a published traffic model, treating inference latency as a first-order criterion, and quantifying seed variability with multiple-comparison control (Table I). The gap is therefore one of evidence on robustness, not of detectors.

Objectives and questions. *General objective:* to determine whether DL detectors provide a reproducible advantage over a rolling-statistics baseline (G0) on PeMS-D4 and whether the detector ranking is stable across labeling schemes. *Specific objectives:* (O1) estimate, under Scheme B, the difference in PR-AUC between each DL detector and G0; (O2) establish which detectors that exceed G0 meet a predefined latency criterion; (O3) quantify the agreement between the detector rankings under Schemes A and B. *Research questions:* RQ1 (O1, H1), by how much, if at all, does any DL detector improve average precision over G0 under Scheme B; RQ2 (O2, H2), do the detectors that improve on G0 meet the latency criterion; RQ3 (O3, H3), is the detector ranking stable across labeling schemes? Input-controlled and per-type analyses test the robustness of these comparisons.

Declared contributions. (1) *Empirical:* evidence on whether the standing of DL detectors against a rolling baseline on freeway flow is robust to the anomaly definition (DL results pending); (2) *methodological:* a reproducible protocol for PeMS-D4 that operationalizes practices distilled from the seed studies, with two separately constructed labeling schemes and an a priori ranking-stability criterion; (3) *practical:* latency as an explicit criterion, with a PR-AUC×latency Pareto front. The preliminary outputs reported here concern the data, the labels and the baseline only.

II. Related Work and Distilled Practices

Traffic anomaly detection. Reported detectors include graph-attention forecasting [4], CNN and BiLSTM autoencoders [5], a GCN-LSTM-attention autoencoder evaluated with artificial anomalies [6], and SI-DMSeq, a multitask LSTM encoder–decoder that flags disruptions with a control chart [7]. Labeling is the main obstacle: weak supervision raised the incident-class F1 of an LSTM from 0.69 to 0.88 on PeMS District 4 [9], and FT-AED combines crash logs with expert labels [8]. On expert-labeled volume data, a mean$\pm c\cdot$std time-of-day profile reached F1 0.566 versus 0.120 for an LSTM [17].

TABLE I

Protocol features reported by seed studies. Tr: traffic data; NB: non-deep baseline; CS: chronological split stated; NT: no PA and no test-tuned threshold for the headline metric; RR: repeated runs; ST: significance test; LT: runtime/latency. ✓ yes, ✗ no, p partial, n.r. not reported, – not applicable, plan: planned, not yet executed.

Study	Tr	NB	CS	NT	RR	ST	LT
Taheri'26 [17]	✓	✓	n.r.	✗	n.r.	✗	✗
Taheri'24 [22]	✓	✓	n.r.	✗	✓	✗	✗
FT-AED'24 [8]	✓	p	n.r.	✓	✗	✗	✗
Sun'22 [9]	✓	✗	✓	✓	✓	✗	✗
Zhang'22 [4]	✓	✓	n.r.	✓	✗	✗	✗
Ijaradar'25 [6]	✓	✓	n.r.	n.r.	n.r.	✗	✗
Nouri'25 [7]	✓	✗	✓	–	✗	✗	✓
Kim'22 [11]	✗	✓	–	✗	p	✗	✗
Audibert'22 [14]	✗	✓	–	✗	n.r.	✓	✗
Schmidl'22 [13]	✗	✓	–	✓	n.r.	✗	✓
Liu'24 [16]	✗	✓	–	✓	✗	✓	n.r.
This protocol	✓	✓	✓	✓	*plan*	*plan*	*plan*

Evaluation pitfalls. Under PA, uniform random scores reach F1 0.969 on SWaT versus 0.216 without it [11]; 316 of 367 Yahoo series are separable by a moving mean/std one-liner [12]. AUC-ROC is dominated by true negatives under imbalance [16], [18], but point-wise PR-AUC is not robust to small temporal lags [16].

Synthetic labels. Anomalies split into point-wise (global, contextual) and pattern-wise types [19]. Injected anomalies help to train and select detectors, but differ from real ones and transfer only when types match [20], [21].

Table I summarizes the protocol features reported by the seed studies and motivates each design choice in Section III.

III. Protocol

Design. Computational comparative experiment with a factorial structure (detector × labeling scheme) and repeated measures over sensor×fold blocks, under walk-forward temporal validation and one held-out test block.

Data. PeMS-D4 [23] contains flow, occupancy and speed from 307 freeway sensors in the San Francisco Bay Area, 1 Jan–28 Feb 2018, at 5-min resolution (16 992 steps per sensor). Sensors with more than 20% zeros or a 24-h frozen run were excluded *a priori*, leaving 291. Each sample is a 12-step window ($t-11..t$, one hour) with $F = 8$ inputs (flow, occupancy, speed, cyclic hour and weekday encodings, a peak-hour flag); the label is that of the final step t, so all methods score the same instant. One global model per architecture is trained on the windows of all sensors.

Partition. The last 20% of each series is a held-out test block used once. The first 80% forms three expanding walk-forward folds; z-score normalization, hyperparameters and thresholds are fitted only on the training and validation parts of each fold, never on test data (Table I, CS/NT).

Labeling schemes. *Scheme A* (proxy) labels $x_{s,t}$ as anomalous when $|z_{s,t}| > 3$, with mean and standard deviation computed causally on the previous 288 steps of raw flow; the anticipated prevalence was 0.3–2%. The baseline G0 belongs to the same rule family (rolling mean and standard deviation, although over a much shorter window), so A is used only for the ranking-stability test (H3), never for H1. *Scheme B* injects point, contextual and collective (3, 6 or 12 steps) anomalies into real series, perturbing flow, occupancy and speed, at a 3% target prevalence per split and per sensor×fold block, with a fixed injection seed, physical constraints (flow, speed ≥ 0; occupancy in $[0, 1]$) and no overlap across partitions. Contextual anomalies replace a value with one below the 5th or above the 95th percentile of a per-sensor hour×weekday profile estimated on training data, a rule family distinct from G0 [19].

Conditions. G0: non-parametric rolling detector on flow only; it scores each step by the absolute z-score over the previous w steps (window $w \in \{6, 11, 22, 44\}$ steps and threshold multiplier $k \in \{2.0, 2.5, 3.0\}$, a 12-point grid). G1: LSTM; G2: GRU; G3: CNN-1D; G4: autoencoder (reconstruction error as score); G5: Transformer encoder; G6: CNN-LSTM-GRU hybrid; G7: SI-DMSeq [7] with its published configuration, using the local reconstruction error of the series at step t as the point score. G1–G3, G5 and G6 optimize a supervised classification loss (sigmoid output, binary cross-entropy). G4 optimizes a reconstruction loss on windows whose final-step label is normal under the corresponding scheme, so those windows may contain earlier anomalous steps. G0 and G7 optimize no supervised classification loss, although label-based selection is used where specified (G0's window and threshold on validation). Architecture and degree of label use are therefore partly confounded, and the comparison is between detector systems rather than between architectures in isolation. G0–G6 are implemented in TensorFlow/Keras. Each of G1–G6 evaluates 12 configurations per scheme, the first being its initial specification and the other 11 drawn from the fixed ranges of Table II with a deterministic seed. G7 keeps PyTorch and no tuning, an asymmetry declared a priori.

Reduced search. The nominal search (6 architectures × 2 schemes × 3 folds × 12 configurations × 3 seeds) is 1 296 trainings; at the 4.3–5.4 min per epoch measured on fold 3 (CPU, LSTM), 30 epochs each exceed 70 days. Before any search we fixed one reduction for all architectures: one seed, at most 10 epochs with early stopping (patience 3) on PR-AUC, every fourth training window, and selection on the *full* validation block. The winner is retrained on the whole 80% block with three seeds, all windows and no early stopping, using the median best epoch. Cost was timed on an AMD Ryzen 7 7445HS (6 cores, 12 threads) with 15.2 GB of RAM, on CPU (TensorFlow 2.16.2); a GPU run (RTX 4050, under WSL2) was slower for this model and batch size (about 12 versus 5.4 min per epoch). Each run records its hardware, and training times are compared across architectures only when measured on the same machine.

Selection and metrics. The configuration maximizing mean validation PR-AUC over the three folds is selected; the F1 threshold and the threshold reaching recall 0.90 are chosen on the last validation block (the final models are retrained on that block, which may bias F1 and FPR but not PR-AUC). The primary metric is PR-AUC (average precision), whose chance level equals the prevalence. Block-mean AP is the unweighted mean of sensor-level AP over evaluable blocks; pooled AP is computed on scores and labels concatenated over all valid steps of a split and is reported with its prevalence. G0's PR-AUC depends on w only; k sets the binary threshold used for F1. F1, AUC-ROC and the false-positive rate (FPR) at the F1 threshold and at the recall-0.90 threshold are secondary, and the recall actually reached on test is reported with that FPR; PA is forbidden [11], [18]. Because collective anomalies dominate the injected instants, AP is also reported per anomaly type, taking the instants of one type as positives and the normal instants as negatives (other types excluded). Latency is measured with `perf_counter` after 100 warm-up calls over ≥1000 calls, at batch 1 and for one window per sensor, reporting median, p95 and 95% CI; the operational criterion is ≤10 ms per window, complemented by the PR-AUC×latency Pareto front.

TABLE II
SEARCH SPACE OF G1–G6. THE FIRST VALUE OF EACH LIST IS THE INITIAL SPECIFICATION (CONFIGURATION 0); THE OTHER 11 CONFIGURATIONS ARE DRAWN FROM THESE LISTS.

Model	Values
G1 LSTM, G2 GRU	units $\{[64, 32], [32, 16], [128, 64]\}$; dropout $\{0.1, 0.2, 0.3\}$; learning rate $\{5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$
G3 CNN-1D	filters $\{[32, 64, 128], [16, 32, 64], [64, 128, 256]\}$; kernel $\{3, 5\}$; weight decay $\{10^{-4}, 0, 10^{-3}\}$; learning rate as G1
G4 autoencoder	layers $\{[32, 16, 8], [64, 32, 16], [32, 16, 4]\}$; L2 $\{0, 10^{-5}, 10^{-4}\}$; learning rate as G1
G5 Transformer	blocks $\{2, 1, 3\}$; heads 4; d_{model} $\{64, 32\}$; feed-forward $\{128, 64\}$; dropout $\{0.1, 0.2\}$; learning rate $\{5 \times 10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$; weight decay $\{10^{-4}, 10^{-3}\}$
G6 CNN-LSTM-GRU	filters $\{64, 32\}$; kernel $\{3, 5\}$; LSTM units $\{64, 32\}$; GRU units $\{32, 16\}$; dropout $\{0.2, 0.1, 0.3\}$; learning rate as G1

Controls against a signature shortcut. Networks see eight channels, G0 only flow, and Scheme B perturbs several channels, so a network could exploit the injection signature instead of detecting flow anomalies better. Per-type AP and an ablation retrained with occupancy and speed zeroed address this; the ablation keeps the hour, weekday and peak-hour channels, so information is still not equal to G0's. Class-weighted training tests sensitivity to imbalance, not the shortcut. These analyses were fixed after a minimal exploratory run that is not reported as a result.

Hypotheses and statistics. H1: at least one DL detector exceeds G0 in block-mean test PR-AUC under Scheme B by at least $\delta = 0.05$ of absolute PR-AUC (about 14% of G0's 0.345), a margin that rules out trivial gains and is fixed before the test block is opened (H0: none does). H2, conditional

on H1: at least one detector satisfying H1 meets the latency criterion (median of one window, batch 1, model inference only, common CPU; p95 also reported); H0: none does. H3: the detector ranking is not stable across schemes (Kendall's $\tau < 0.7$); H0: $\tau \geq 0.7$. Each alternative states the effect to be demonstrated (an advantage, a latency pass, a ranking change), so H3 is phrased as instability. Validation folds serve selection and exploratory comparisons (Friedman test on the 291×3 blocks, Nemenyi [24]); H1 is evaluated primarily on the 291 held-out test blocks, after configurations, seeds, aggregation rules and comparisons are frozen, with paired tests against G0 and Holm correction ($\alpha = 0.05$). Sensors and expanding folds are not independent and Holm does not correct dependence, so significance is indicative and effect sizes and intervals are reported. Block AP is computed per seed and averaged over the three seeds before testing; per-seed results are also reported. Blocks lacking either class are excluded and reported. For H3 the conditions are ranked by block-mean PR-AUC per scheme (blocks valid under both, resampled for a 95% bootstrap CI of τ): the ranking is stable if the whole CI is at or above 0.7, H3 is supported if it lies below, and the result is inconclusive otherwise; Spearman's ρ is a sensitivity check. As H3 changes labels, prevalence, training examples, selected configurations and perturbed data together, it assesses the sensitivity of the whole procedure to the anomaly definition, not of the evaluation labels alone. Under Scheme A the advantage over G0 is reported descriptively only, because A shares G0's rule family.

IV. Preliminary Results

Table III reports the evidence obtained so far (weeks 1–8) and what remains pending. The pipeline, both labeling schemes and G0 passed 50 of 50 automated checks, including perturbation tests that alter future values; the window loader passed 73 of 73 and the model definitions 54 of 54 checks against the initial specification.

Four observations shape the remaining work. First, the proxy and injected labels are almost disjoint (Jaccard ≤ 0.0012), so H3 is not trivially satisfied. Second, Scheme A has a prevalence below the anticipated 0.3–2% and only 29.2% of blocks with both classes, which limits the block intersection on which H3 is computed; its two aggregations also disagree for G0 (0.084 block-mean over 84 evaluable blocks, 0.003 pooled, about three times the pooled prevalence), so every PR-AUC is reported with its aggregation. Third, G0 is far above chance under injection, consistent with the competitiveness of simple detectors [13], [17], but strongly type-dependent: it separates point anomalies almost perfectly and collective ones poorly, and collective anomalies are three quarters of the injected instants, so any DL gain must be read per type. Fourth, the measured training cost (4.3–5.4 min per epoch on CPU) forced the reduced search of Section III, only partly mitigated by multi-seed retraining.

TABLE III
Evidence obtained to date versus pending analyses. Chance level equals prevalence.

Item	Result
Obtained	
Sensors retained after QC	291 of 307 (16 excluded, 5.2%)
Scheme A prevalence	0.10% (4 873 / 4 860 864); 0.086–0.125% per partition
Scheme B prevalence	3.02–3.06% per block
Scheme B composition (instants)	74.7% collective, 14.6% point, 10.7% contextual
Overlap A vs. B (Jaccard)	0.0003–0.0012
Blocks with both classes	A: 340/1 164 (29.2%); B: 100%
G0 selected config. (A and B)	window 22, multiplier 3.0 (multiplier at grid edge; it does not affect PR-AUC)
G0 test, Scheme B	block-mean PR-AUC 0.345 ($\approx$11.3$\times$ chance); pooled 0.335 (11.0$\times$); F1 0.323; AUC-ROC 0.792; FPR 54.3% at the threshold reaching recall 0.90 on validation
G0 test, Scheme B, per type	point 0.905; contextual 0.399; collective 0.131
G0 test, Scheme A	block-mean PR-AUC 0.084 (84 of 291 blocks evaluable); pooled 0.003 (3.0$\times$); F1 0.021; FPR 65.6% at the validation-derived recall-0.90 threshold
Pending	
G1–G6 search and final training	*pending*
Friedman/Holm, Nemenyi (H1)	*pending*
Latency, Pareto front (H2); G7 smoke test	*pending*
Kendall τ, Spearman ρ (H3)	*pending*
Channel ablation, class weights	*pending*

V. Discussion and Threats to Validity

Prior evidence that statistical baselines can rival DL [14], [17] motivates H1, but failure to detect a DL advantage will be reported with effect sizes and uncertainty, without interpreting non-significance as evidence of equivalence. The published latency of SI-DMSeq (about 55 ms per sequence for 46 zones and 24-step windows [7]) refers to a different load, so all methods will be measured under one configuration; the 10 ms criterion (about 3 s for one window per sensor, far below the 5-min sampling interval) is a technical criterion of this study.

External validity: one dataset, one freeway network and about eleven test days limit generalization; H1 concerns injected anomalies, not verified incidents. *Construct validity:* neither scheme represents verified incidents; injected anomalies differ from real ones [20], and unlabeled real events act as pseudo-negatives [21]. *Internal validity:* Scheme A shares its rule family with G0, hence its restricted role; all fitting is fold-local. G0 uses flow only and a 22-step window against 12 steps and eight channels for the networks, and the reduced one-seed search may favor architectures that converge early. The shortcut controls were fixed after a small exploratory run. *Conclusion validity:* expanding folds are not independent and sensors share dynamics, so the validation-fold Friedman analysis is exploratory and a fold-3-only analysis is planned as a robustness check. *Metric validity:* point-wise PR-AUC weighs long (collective) anomalies more and is lag-sensitive

[16], [18]; per-type results will be reported. *Implementation validity:* G7 runs in a different framework, a declared confounder for latency; training and latency hardware will be documented separately.

VI. Conclusion

We specified and partially executed a uniform, reproducible protocol for comparing DL detectors with a non-parametric baseline on PeMS-D4 freeway flow under two separately constructed labeling schemes. Preliminary outputs show that the schemes yield largely non-overlapping positive sets, that the baseline is well above chance under injection yet type-dependent, and that the nominal tuning budget was infeasible, leading to a declared reduced search. Next come the DL battery, tests for H1–H3, shortcut controls and the latency benchmark; code (MIT license), configurations and seeds will be released with the results.

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